Cyclistic Bike-Share Data Analysis Project

Data Analyst: Rifaldi Adi Putra
Client/Sponsor: Cyclistic Marketing Team

Purpose:

Analyze Cyclistic's historical bike-trip data to understand how annual members and casual riders use the bike-share service differently. The findings support marketing strategies to increase annual memberships.

Scope / Major Project Activities:

Activity	Description
Data Collection	Gather Cyclistic historical bike-trip data from Divvy Bike's public dataset.
Data Cleaning	Inspect data quality, remove duplicates, handle missing values, standardize data types, and prepare the dataset for analysis.
Data Transformation	Create additional variables such as ride duration, day of week, month, and hour to support analysis.
Exploratory Data Analysis	Compare riding behavior between annual members and casual riders using descriptive statistics and trend analysis.
Data Visualization	Develop visualizations to illustrate ride frequency, trip duration, seasonal trends, weekday usage, and bike preferences.
Business Insights & Recommendations	Summarize key findings and provide actionable recommendations to support membership growth strategies.

Final Report	Compile the complete analysis, methodology, visualizations, findings, and recommendations into a final report.

This project does not include:

- Developing or implementing marketing campaigns.
- Predictive modeling or machine learning.
- Analysis outside the provided historical bike-trip dataset.
- Financial analysis or cost-benefit calculations for marketing initiatives.
- Evaluation of post-implementation marketing performance.

Deliverables:

Deliverable	Description/ Details
Cleaned Dataset	A cleaned and transformed dataset ready for analysis.
Exploratory Data Analysis	Summary statistics comparing casual riders and annual members.
Data Visualizations	Charts and graphs illustrating customer riding behavior and usage patterns.
Business Insights	Key findings explaining behavioral differences between rider groups.
Recommendations	Three data-driven recommendations to encourage casual riders to purchase annual memberships.
Final Report	A comprehensive report documenting the business task, data sources, data cleaning process, analysis, visualizations, findings, and recommendations.

Schedule Overview / Major Milestones:

Milestone	Expected Completion Date	Description/Details
Business Understanding	14 July 2026	Define business objective and analytical questions.
Data Collection	15 July 2026	Obtain Cyclistic historical trip data.

Data Cleaning & Preperation	16 July 2026	Clean, validate, and transform the dataset.
Exploratory Data Analysis	17 July 2026	Analyze riding behavior differences between customer types.
Data Visualization	18 July 2026	Create charts and dashboards to communicate findings.
Business Recommendations	19 July 2026	Develop recommendations based on analytical insights.
Final Report	20 July 2026	Complete documentation and present project results.

*Estimated date for completion:

Project Duration: 7 Days